

USABILITY STUDY REPORT

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Recordings link:

https://drive.google.com/drive/folders/1C3aInl8lf65OgOCZdIGtJtLVFJHQ06i?usp=drive_link

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1. Introduction

This usability study evaluates **SignLingo**, a web-based American Sign Language (ASL) learning platform designed to support beginners through structured lessons, quizzes, live practice, and progress-tracking features.

The purpose of this study is to:

- Assess how effectively new users can understand, navigate, and use the system.
- Identify barriers impacting learning, engagement, or task completion.
- Provide data-driven recommendations for improving usability, accessibility, and overall user experience.

This evaluation was conducted with **three representative participants**, all interacting with a functional prototype of SignLingo.

2. Application Summary

SignLingo enables learners to progress through ASL content in a guided, gamified format. Key functionalities include:

- **Structured lessons** with sign-by-sign learning
- **Timed quizzes** with scoring, stars, and level progression
- **Live practice** using webcam detection (optional for learners)
- **Character-based sign browsing** with search
- **Profile dashboard** tracking XP, level, streaks, and lesson/quiz status

The system is built using **React, Node.js, Express, and MongoDB**, with a session-based authentication model.

3. Target Audience & Personas

The target users include:

1. **Beginner learners** with no prior ASL experience
2. **Family members or caregivers** of Deaf/HoH individuals
3. **Students, volunteers, or community members** requiring quick-access sign learning

4. Young adults comfortable with web learning platforms

User Personas:

Persona 1 — Maya Lopez (Beginner ASL Learner)

Age: 21

Occupation: College student

Technical Comfort: High

Goals:

- Learn foundational ASL quickly for personal interest.
- Track daily progress and stay motivated with streaks and XP.
- Access short, visual lessons and quizzes for bite-sized learning.

Frustrations:

- Overwhelmed by long, text-heavy ASL resources.
- Unsure how to measure or validate her progress.

Behaviors:

- Uses laptop and mobile during commute.
- Prefers gamified learning and quick interactions.
- Relies on intuitive navigation with clear visual hierarchy.

Why SignLingo helps:

Short lessons, quizzes, XP, and a leaderboard keep her engaged and progressing.

Persona 2 — Ravi Patel (Family Member / Caregiver)

Age: 38

Occupation: Retail Manager

Technical Comfort: Medium

Goals:

- Learn common everyday signs to communicate with his Deaf niece.
- Needs guided path without confusion or multiple steps.
- Desires clear instructions and reassurance that he is signing correctly.

Frustrations:

- Most ASL resources online are unstructured and visually unclear.
- Hard to know if he's performing signs accurately.

Behaviors:

- Prefers desktop usage in the evening.

- Reads step-by-step explanations and relies heavily on visuals.
- Needs simple buttons, high contrast, and accessible UI.

Why SignLingo helps:

Structured lessons, large sign images, and live practice build confidence.

Persona 3 — Sarah O’Brien (Community Volunteer / Student Ambassador)

Age: 24

Occupation: University volunteer working with Deaf/HoH groups

Technical Comfort: High

Goals:

- Learn practical ASL phrases used in events and community interactions.
- Wants fast searching and quick reference to signs.
- Needs to refresh gestures right before events.

Frustrations:

- Difficult to navigate cluttered sign libraries.
- Slow-loading videos disrupt her practice.

Behaviors:

- Uses a laptop during planning meetings.
- Frequently accesses the Characters/Sign gallery.
- Uses keyboard navigation during work sessions.

Why SignLingo helps:

Fast search, clear categories, and keyboard accessibility streamline her workflow.

Persona 4 — Ethan Miller (Young Adult Digital Learner)

Age: 17

Occupation: High school student

Technical Comfort: Very High

Goals:

- Explore ASL out of curiosity and for college applications.
- Wants fun experiences like quizzes and leaderboards.
- Enjoys competing with friends.

Frustrations:

- Gets bored easily if the interface feels plain or slow.
- Dislikes apps with unclear navigation or long text.

Behaviors:

- Uses mobile/laptop interchangeably.
- Responds well to gamification (XP, streaks, stars).
- Skips around and explores different sections quickly.

Why SignLingo helps:

Gamified experience, XP rewards, and quizzes maintain engagement.

4. Data Observed in the System

The system processes and displays:

- **User information:** name, age, email, authentication credentials
- **Lesson progress:** completed signs, timestamps, XP earned
- **Quiz data:** session state, results, accuracy, stars earned
- **Sign database:** gloss, description, category, media
- **Leaderboard metrics:** aggregated XP and level

This data allows evaluation of user comprehension, progress visualization, and decision-making within the interface.

5. Key User Tasks (Use Cases)

To evaluate real-world usage, the following tasks were selected:

T0 - Initial Understanding

Can the user understand the purpose of the application from the flashscreen alone?

T1 - Authentication

Log in or sign up.

Measure: Visibility of sign-up option, clarity of login flow, error messages.

T2 - Start a Lesson

Navigate to *Lessons*, open a lesson, and begin learning.

Measure: Discoverability of buttons, comprehension of lesson layout.

T3 - Complete a Quiz

Start and complete at least three questions.

Measure: Interpretation of correctness feedback, quiz navigation.

T4 - Sign Search

Search for any sign in the Characters page.

Measure: Recognition of search field and understanding of results.

T5 - Profile Interpretation

Access the profile page and explain editable user information.

Measure: Clarity of profile layout, understanding of fields, confidence in editing.

6. Experiment Procedure

6.1 Preparation

All sessions were recorded via Zoom. Participants interacted with the web app in a full-screen browser environment.

6.2 Introduction Script

Participants were greeted with a standardized script describing:

- Purpose of the study
- Think-aloud method
- Voluntary participation
- Recording consent

All participants **verbally consented**.

6.3 Demographics Collected

- Age
- Familiarity with ASL
- Familiarity with learning apps
- General technical comfort

6.4 Methodology

Participants were guided through the task script (T0–T5) while narrating their thought processes. Observers captured errors, delays, confusions, and successful behaviors.

A **post-test Likert scale questionnaire** was used to gather subjective satisfaction scores for each task and overall usability.

7. Detailed Findings by Participant

Participant 1 - Nithin (Age 25)

Observations

- Correctly interpreted flash screen purpose.
- Attempted to sign up from the Login form due to low visibility of the Sign-Up link.

- Noted slow login feedback (perceived as glitch).
- Found Lesson Start and Reset buttons insufficiently prominent.
- Quiz feedback was understood but not visually strong.
- Did not realize Characters images were expandable.

Notable Quotes

- “I didn’t even see the Sign-Up option.”
- “Is the login stuck?”

Participant 2 - Vandana (Age 26)

Observations

- Understood the app immediately.
- Same Sign-Up visibility issue as P1.
- Significant difficulty locating Start Lesson and Reset buttons.
- Believed correct answer feedback was missing (it existed but lacked visibility).
- Did not realize Characters cards expand.
- “View All” label caused confusion.

Notable Quotes

- “I don’t think the correct answer is shown... oh wait, it is too far down.”

Participant 3 - Suhail (Age 24)

Observations

- Instantly recognized the purpose of the application.
- Login flow was smooth; requested Forgot Password.
- Lesson layout intuitive but Start button remained difficult to spot.
- Quiz feedback worked well for him—contradictory experience to others.
- Characters cards small but acceptable.
- Significant uncertainty regarding editable Profile fields.

Notable Quotes

- “Why is email editable? Isn’t that unsafe?”

8. Likert Scale:

1 = Strongly Disagree

2 = Disagree

3 = Neutral

4 = Agree

5 = Strongly Agree

Question	Participant 1	Participant 2	Participant 3
Q1 – Easy to navigate	5	5	5
Q2 – Tasks easy	4	4	5
Q3 – Dashboard clear	4	5	5
Q4 – Quiz instructions clear	5	5	5
Q5 – Feedback helpful	4	4	5
Q6 – Characters search intuitive	5	5	5
Q7 – Would use again	5	5	5

Question	Mean Score	Interpretation
Q1 – Easy to navigate	5.0	Excellent
Q2 – Tasks easy	4.33	Very Good
Q3 – Dashboard clear	4.67	Excellent
Q4 – Quiz instructions clear	5.0	Excellent
Q5 – Feedback helpful	4.33	Very Good
Q6 – Characters search intuitive	5.0	Excellent
Q7 – Would use again	5.0	Excellent

9. Comparative Analysis (Across Participants)

Strengths

- Purpose of the app universally understood (T0)
- Navigation structure intuitive
- Users enjoyed quizzes and felt engaged
- Profile page was easy to reach and visually clean

Recurring Issues

1. **Sign-Up entry point too subtle**
2. **Lesson Start and Reset buttons were consistently overlooked**
3. **Incorrect/correct feedback in quizzes not visually emphasized**
4. **Characters images too small; affordance for expansion unclear**
5. **Email editability raised security concerns**
6. **Ambiguous “View All” label in Profile**
7. **Login session feedback delayed or unclear**

10. Prioritized Recommendations

Priority	Issue	Recommendation	Status
Must Fix	Sign-Up visibility	Increase contrast, reposition CTA	Implemented
Must Fix	Authentication clarity	Lock email field	Implemented
Must Fix	Routing/session bugs	Stabilize refresh/logout behavior	Implemented
Should Fix	Lesson button visibility	Larger button, repositioning, color emphasis	Not implemented
Should Fix	Quiz feedback subtlety	Highlight correct answer, bring closer to question	Implemented
Could Fix	Characters thumbnails small	Increase size, add hover state, add “Click to enlarge”	Partially implemented
Could Fix	“View All” ambiguity	Provide descriptive labels	Not implemented

11. Conclusion

The usability study confirms that **SignLingo has a strong conceptual design**, intuitive navigation, and engaging interactive learning components. Participants clearly understood the purpose of the platform and enjoyed core features such as lessons, quizzes, and sign browsing.

However, the study also revealed **several visibility and affordance issues**, all of which directly influenced learnability. These findings informed targeted improvements implemented in Project 4, significantly improving authentication clarity, visual hierarchy, button discoverability, feedback prominence, and overall accessibility.

SignLingo now supports a smoother, more accessible, and more trustworthy onboarding and learning experience.